# AIMCR FORENSIC REPORT

AI-Powered Mobile Forensic Intelligence Platform for Autonomous Crime Reconstruction

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|---|---|
| **Case Reference:** | crime2 |
| **Case Status:** | Active |
| **Date Generated:** | 2026-08-05 09:30:34 |
| **Assigned Investigator:** | Lead Investigator John |
| **Evidence Count:** | 30 artifacts ingested |
| **Forensic Complexity Index:** | 80/100 (Critical) |

## Executive Summary

This document compiles explainable investigative findings extracted from mobile device artifacts using the Autonomous Crime Reconstruction and Behavioral Profiling engines. Every finding detailed within this report is traceably backed by a corresponding SHA-256 hashed forensic artifact, maintaining full chain of custody. The contents represent investigative directions and indicators of probability rather than definitive judicial outcomes.

# 1. Incident Reconstruction

No structured cybercrime sequences were reconstructed by the engine.

# 2. Behavioral Anomaly Log

The Isolation Forest engine analyzed the user's historical feature vectors (active hours, GPS coordinates, application categories, file extensions) and highlighted the following deviations from baseline behavior:

| Timestamp | Type | Score | Anomaly Explanation / Context | Ref |
|---|---|---|---|---|
| 2026-08-05 08:25:00 | GPS | 1.00 | Flagged location anomaly at coordinates (34.0522, -118.2437). This location is far outside the user's standard historical geographic cluster. | 0124192d |
| 2026-08-05 03:58:40 | DeviceInformation | 0.85 | Behavioral baseline deviation detected based on multi-variate event clustering. | 785f6371 |
| 2026-08-05 08:10:00 | GPS | 0.76 | Flagged location anomaly at coordinates (40.7128, -74.006). This location is far outside the user's standard historical geographic cluster. | 87d47f4c |
| 2026-08-05 08:20:00 | GPS | 0.76 | Flagged location anomaly at coordinates (40.713, -74.0058). This location is far outside the user's standard historical geographic cluster. | 6df8bdb5 |

# 2a. Anti-Forensics & Tampering Log

Heuristic scanners inspected the integrity of the device artifacts to find evidence of active concealment, extension spoofing, file system timestamp manipulation, or database tampering:

No active anti-forensics attempts or database anomalies were flagged.

# 3. Relationship Network & Priority Entities

The communication graph maps contacts, browser domains, and apps into connected modular communities. Nodes are assigned an Investigative Priority score using degree and betweenness centralities.

### Priority Investigation Targets

| Entity Identifier | Node Class | Investigative Priority Score | Action Needed |
|---|---|---|---|
| secure-auth-bank.net | Location | 85.0/100 | Immediate Forensics Review |
| GPS Location Logger | Application | 22.67/100 | Monitor & Validate |
| Coords: 40.713, -74.006 | Location | 10.0/100 | Monitor & Validate |
| Chrome Browser | Application | 10.0/100 | Monitor & Validate |
| Coords: 34.052, -118.244 | Location | 10.0/100 | Monitor & Validate |
| Bank Update Security | Application | 0.0/100 | Monitor & Validate |

# 3a. Forensic Investigation Recommendations

**AI/Deterministic Recommendations Summary:** Based on the analysis of active threat intelligence matches and anti-forensics indicators, the investigator should prioritize disassembling the spoofed APK binary (disguised as a PNG) and carving the tampered SQLite database. Reconstructing network infrastructure and verifying modification timestamps via system logs is critical to confirm the attack vector and trace perpetrator activity.

| Recommended Actions | Priority | Forensic Rationale / Next Steps |
|---|---|---|
| Perform WHOIS and passive DNS resolution query on 'secure-auth-bank.net' | High | The phishing infrastructure domain/URL 'secure-auth-bank.net' was matched. This will identify domain registrars, hosting IPs, and subdomains. |
| Check proxy/DNS resolution logs on client device for contacts with 'secure-auth-bank.net' | Medium | Determines exact duration and timeline of client communication with the malicious infrastructure. |

# 4. Evidence Integrity Index (Chain of Custody)

Every artifact imported into the platform is indexed below with its original acquisition timestamp and a SHA-256 integrity hash to ensure transparency and auditability.

| Evidence ID | Timestamp | Type | Source File | SHA-256 Hash Signature |
|---|---|---|---|---|
| e7b8d9fa... | 2026-08-05 03:58:40 | Contact | contacts.json | 8e261bede328f224... |
| 95d909c0... | 2026-08-05 03:58:40 | Contact | contacts.json | 8e261bede328f224... |
| f6b9ecb8... | 2026-08-05 03:58:40 | Contact | contacts.json | 8e261bede328f224... |

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|---|---|---|---|---|
| 785f6371... | 2026-08-05 03:58:40 | DeviceInformation | device_info.json | 148a8f2cfc0c2115... |
| 2ba16594... | 2026-08-05 03:58:40 | BrowserHistory | wifi_history.json | 1b9a61588b329133... |
| 50b6c269... | 2026-08-05 03:58:40 | BrowserHistory | wifi_history.json | 1b9a61588b329133... |
| e6194c4d... | 2026-08-05 07:30:00 | Call | calllogs.json | 1f418020913b7d51... |
| 3a6d15e6... | 2026-08-05 08:00:00 | SMS | sms_messages.json | c635e3c698d77702... |
| f0359c95... | 2026-08-05 08:05:00 | SMS | sms_messages.json | c635e3c698d77702... |
| 87d47f4c... | 2026-08-05 08:10:00 | GPS | gps_locations.json | e0b1cda71b7ed6f3... |
| bfd4b8ee... | 2026-08-05 08:15:00 | TimelineEvent | case_timeline.json | b9d3c4f82bbf775d... |
| b8868f2b... | 2026-08-05 08:15:00 | SMS | sms_messages.json | c635e3c698d77702... |
| 19f9e47a... | 2026-08-05 08:15:05 | Notification | notifications_log.json | 6c79e54c25058488... |
| 7034f199... | 2026-08-05 08:16:00 | Call | calllogs.json | 1f418020913b7d51... |
| d211f9f8... | 2026-08-05 08:16:30 | TimelineEvent | case_timeline.json | b9d3c4f82bbf775d... |
| fd55cb71... | 2026-08-05 08:16:30 | BrowserHistory | chrome_history.json | d8c4bdf4e19179ff... |
| ae1d1a3e... | 2026-08-05 08:17:00 | Bluetooth | bluetooth_devices.json | 3b73a9f4263dd850... |
| 1c6bcda9... | 2026-08-05 08:18:00 | BrowserHistory | chrome_history.json | d8c4bdf4e19179ff... |
| a9e88c18... | 2026-08-05 08:19:00 | TimelineEvent | case_timeline.json | b9d3c4f82bbf775d... |
| 827545a6... | 2026-08-05 08:19:00 | Application | installed_apps.json | 73e6b31fbd09f62b... |
| 6df8bdb5... | 2026-08-05 08:20:00 | GPS | gps_locations.json | e0b1cda71b7ed6f3... |
| 626eebf8... | 2026-08-05 08:20:30 | Application | installed_apps.json | 73e6b31fbd09f62b... |
| 96a174ac... | 2026-08-05 08:21:00 | TimelineEvent | case_timeline.json | b9d3c4f82bbf775d... |
| 14911dc7... | 2026-08-05 08:21:00 | SMS | sms_messages.json | c635e3c698d77702... |
| 89fe9568... | 2026-08-05 08:21:05 | Notification | notifications_log.json | 6c79e54c25058488... |

** Showing 25 of 30 total evidence records. Full trace available in database.*

## 5. System Investigation Audit Trail

Immutable audit log records detailing investigator queries, report configurations, and analyses run inside this case file:

| Timestamp | Action | Investigator details / Queries |
|---|---|---|
| 2026-08-05 03:59:58 | QUERY_COPILOT | Asked Copilot: 'Summarize this investigation case.'. Response cited 5 artifacts. |
| 2026-08-05 03:59:24 | RUN_ANALYSIS | Executed AI behavioral profiling & communication graph analysis for case ID: 94b3b59a-aefc-4dcd-bc20-79709aabfaa6. Detected 4 anomalies and 2 Threat Intel match(es). |

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|---|---|---|
| 2026-08-05 03:58:40 | IMPORT_EVIDENCE | Imported forensic file/archive: mock_case_evidence.zip to case ID: 94b3b59a-aefc-4dcd-bc20-79709aabfaa6. Success: 30, Skipped: 0 |
| 2026-08-05 03:57:25 | CREATE_CASE | Created case 'crime2' (ID: 94b3b59a-aefc-4dcd-bc20-79709aabfaa6, Priority: Medium) |
| 2026-08-05 03:57:04 | ACQUIRE_DEVICE_SUCCESS | Successfully acquired device: pixel_6_mock_001. Imported 21 records. |
| 2026-08-05 03:57:02 | ACQUIRE_DEVICE | Initiated forensic acquisition for device: pixel_6_mock_001 in case ID: 4fc01826-208f-4559-999e-e9442d1cca99 |
| 2026-08-05 03:56:30 | LOGIN | User investigator@aimcr.gov logged in successfully |
| 2026-07-24 04:31:57 | LOGIN | User investigator@aimcr.gov logged in successfully |
| 2026-07-24 04:31:45 | LOGIN | User investigator@aimcr.gov logged in successfully |
| 2026-07-24 03:31:46 | EXPORT_REPORT | Generated PDF Investigation Report for Case 'Case #2026-089A: Phishing & Wallet Takeover' (Path: C:\Users\Admin\Documents\ccf\backend\reports\c1dbed0f-3a52-40f3-ae5f-704e625296ff.pdf) |